

Problem

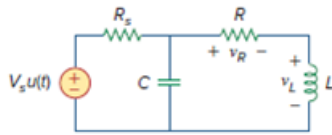
In the circuit of Fig. 8.67, find:

(a) $v_R(0^+)$ and $v_L(0^+)$,

(b) $dv_R(0^+)/dt$ and $dv_L(0^+)/dt$,

(c) $v_R(\infty)$ and $v_L(\infty)$.

Figure 8.67:



Step-by-step solution

Step 1 of 3

(a)

$$\text{At } t < 0, V_s u(t) = 0 \text{ V}$$

Hence current through the inductor is zero.

Therefore,

$$i_L(0^-) = 0 \text{ A}$$

$$v_C(0^-) = 0 \text{ V}$$

Since, the inductor current and capacitor voltage cannot change abruptly. Hence,

$$i_L(0^+) = 0 \text{ A}$$

$$v_C(0^+) = 0 \text{ V}$$

Therefore, the voltage across resistor at $t < 0$ is $v_R(0^+) = 0 \text{ V}$ and the voltage across inductor at $t < 0$ is $v_L(0^+) = 0 \text{ V}$.

Step 2 of 3

(b)

At $t > 0$,

$$V_s u(t) = V_s \text{ V}$$

Since at $t = 0^+$ current flowing through the inductor and resistor is zero and inductor acts like an open circuit, there is no voltage drop across resistor. The current flows through the capacitor is,

$$i_c(0^+) = \frac{V_s}{R_s}$$

$$\text{Also } i_c(0^+) = C \frac{dv_c(0^+)}{dt}$$

$$\text{i.e., } \frac{dv_c(0^+)}{dt} = \frac{V_s}{CR_s}$$

$$v_c(0^+) = v_R(0^+) + v_L(0^+) \text{ (Parallel connected branches).}$$

$$\text{Since, } v_R(0^+) = 0$$

$$\therefore v_c(0^+) = +v_L(0^+)$$

Differentiate both sides,

$$\frac{d(v_c(0^+))}{dt} = \frac{d(v_L(0^+))}{dt}$$

$$\frac{d(v_L(0^+))}{dt} = \frac{V_s}{CR_s}$$

Therefore, the required value is $\frac{dv_L(0^+)}{dt} = \frac{V_s}{CR_s}$ and this leads to $\frac{dv_R(0^+)}{dt} = 0$.

Step 3 of 3

(c)

As $t \rightarrow \infty$, the circuit reaches steady state the inductor will be short circuited and the capacitor will be open circuited.

The voltage across resistor is,

$$v_R(\infty) = V_s u(t) \times \frac{R}{R + R_s} \text{ V}$$

Therefore, the voltage across resistor is $v_R(\infty) = V_s u(t) \times \frac{R}{R + R_s} \text{ V}$.

Since the inductor is short circuited,

$$v_L(\infty) = 0 \text{ V}$$

Therefore, the inductor voltage is $v_L(\infty) = 0 \text{ V}$.